

JEE MAIN REVISION MASTER: PARABOLA

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 3 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Parabola: Standard Forms, Focus & Directrix	3 PYQs	Conics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Parabola: Standard Forms, Focus & Directrix	$y^2 = 4ax \implies \text{Focus: } (a, 0), \text{ Directrix: } x = -a, \text{ Latus Rectum: } 4a, c = \frac{a}{m}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Parabola: Standard Forms, Focus & Directrix (Moderate Yield — 3 PYQs)

Core Principle: Standard parabola $y^2 = 4ax$: Eccentricity $e = 1$. Vertex $(0, 0)$, Focus $(a, 0)$, Directrix $x = -a$, Axis $y = 0$, Length of latus rectum $= 4a$. Condition for line $y = mx + c$ to be tangent to $y^2 = 4ax$: $c = a/m$.

■ **Caution / JEE Pitfall:** For $x^2 = 4ay$, focus is on the y-axis at $(0, a)$ and directrix is horizontal $y = -a$.

■ **Speed Trick:** Focal distance of any point $P(x_1, y_1)$ on parabola $y^2 = 4ax$ is simply $x_1 + a$.

Representative Source PYQs:

- [2024 Shift] Q#1 (MCQ): "The equation of parabola whose focus is $(6,0)$ and directrix is $x=-6$ is..." [Ans: N/A]
- [2021 Shift] Q#2 (MCQ): "If the parabola $y = \alpha x^2 - 6x + \beta$ passes through the point $(0,2)$ and has its tangent at $x=(3/2)$ parallel to X-axis, the..." [Ans: N/A]
- [2020 Shift] Q#3 (MCQ): "If the parabola $x^2 = 4ay$ passes through the point $(2,1)$, then the length of the latus rectum is..." [Ans: B]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2024 (Q#1)

The equation of parabola whose focus is (6,0) and directrix is $x=-6$ is

- (A) $y^2=24x$
- (B) $y^2=-24x$
- (C) $x^2=24y$
- (D) $x^2=-24y$

Correct Answer: None

Solution Analysis: ■ Focus has positive x -coordinate. So, coordinate of focus $= (a,0) \equiv (6,0) \therefore a=6 \dots$ (i) Hence, equation of parabola is $y^2=24x$ [■ $y^2=4ax$ and using Eq. (i)]